

EXERCISES AND EXACT SOLUTIONS

Rainfall and Drought Impact Monitoring

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C13
Notebook	V7_A_N08
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Add a standardized precipitation index with a documented reference period.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Compare station-only and satellite-assisted results.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Add crop-specific calendars.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 4

Explain why a drought alert cannot determine compensation.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Fit/standardize precipitation over an approved long reference series and report distribution assumptions.

Solution 2

Evaluate bias, coverage, spatial representativeness, and disagreement; route large disagreement to verification.

Solution 3

Join locally validated crop-stage dates and propagate calendar uncertainty.

Solution 4

Compensation also requires legal criteria, verified losses, exposure, eligibility, budgets, appeals, and authorized decisions.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated